Grade 10 Test 18 Oct 2023

Electrostatics, magnetism + Electrostatics

MEMO

44

Question One.

1.1. B ✓
1.2. B C ✓✓
1.3. A ✓✓

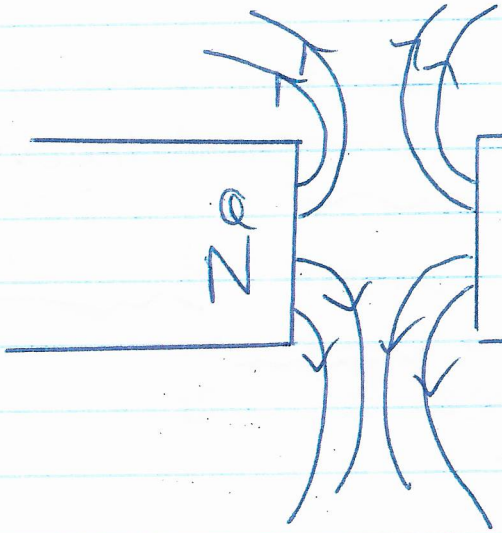
(2x3 = 6)

Question 2

2.1. A SOUTH ✓
P SOUTH ✓
X NORTH ✓

(3)

2.2.



diamond shape ✓
arrows ✓
Not touching ✓

(3)

Coe. Q2.1.

2.3. a) 2 identical magnets ✓✓
b) lose magnetism

(2)

Question 3

(2)

3.1. True. ~~The~~ A compass needle points to magnetic N as it is attracted to S pole of earth's bar magnet. (4)

3.2. Fe_2O_3 is ferromagnetic. In Earth's magnetic field the domains would be aligned ∴ bird could sense direction of earth's magnetic field (3)

Question 4

4.1. Charge (electrons) builds up on car due to friction. Touch door ~~at~~ knob the charge is earthed / discharged / neutral. (3)

4.2. Charged ruler causes polarisation (& explain) in paper leaving surface oppositely charged to ruler ∴ attract. (3)

Question 5

5.1 Charge is not created or destroyed 2
no. of electrons gained = no. lost

5.2. Electrons move from B to A ✓
Both spheres have same charge ✓
∴ Repel. (2)

5.3. $Q_{new} = \frac{+2+(-8)}{2}$ ✓

$= (-3\ nC)$ ✓ (2)

5.4. $n_e = \frac{\Delta Q}{q_e}$

$= \frac{(3 \times 10^{-9}) - (-8) \times 10^{-9}}{1,6 \times 10^{-19}} = \frac{5 \times 10^{-9}}{1,6 \times 10^{-19}}$ ✓m ✓Conv.

$= 3,125 \times 10^{10}$ ✓ (3)

Question 6

6.1. Parallel ✓ (1)

6.2. $\frac{1}{R_p} = \frac{1}{160} + \frac{1}{240}$ OR prod/sum

$\frac{1}{R_p} = \frac{1}{96}$ $= \frac{160 \times 240}{160 + 240}$ ✓

$R_p = 96\ \Omega$ $= 96\ \Omega$ ✓ (2)

(4)

5.3. $I = \frac{12}{96}$ ✓ cao.

$= 0.125\,A$ ✓ (2)

5.4. $R_S = 160 + 240$ ✓

$= 400\,\Omega$ ✓ (2)